llmXive Automated Review of Accurate, Interdisciplinary and Transparent Structure-property Understanding with Deep Native Structural Reasoning

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several high-impact claims regarding state-of-the-art performance and reasoning capabilities across biology, chemistry, and materials science. While the internal logic of the proposed method is sound, there are significant issues with the factual accuracy of the cited baselines and the traceability of specific quantitative claims.

The most critical issue is the citation of non-existent models. The paper repeatedly compares SciReasoner against “GPT-5.5”, “Opus-4.7”, “DeepSeek-V4-Pro”, and “Kimi-K2.6” (Sections 1, 2, and Tables 1-3). The bibliography lists these with 2026 dates (e.g., `openai2026gpt55`, `opus47`). As of the current date, these models do not exist in the public record. Comparing a new model against hallucinated or future-dated baselines invalidates the “state-of-the-art” claim, as the performance gap cannot be verified or reproduced. This is a fundamental failure of claim accuracy regarding the evidence provided.

Additionally, several specific quantitative assertions lack direct support in the provided tables or figures. The abstract and introduction mention specific aggregate statistics (e.g., “82%” or specific ratios) that are not explicitly broken down in the results tables, which list 86 tasks and 67 SOTA wins but do not show the raw numbers to derive a specific “82%” figure if that is what was intended. The claim of “5/5” recovery in retrosynthesis cases versus “2/5” for RSGPT is a strong specific claim, but the specific 5 cases are not enumerated in the text, and the aggregate table only shows a 0.63 Exact Match for RSGPT, making it impossible to verify the 2/5 figure without the raw data. Similarly, the claim that the model’s R^2 of 0.895 is “well above” competing models is not fully supported by Table 1, which reports MAD/MAE ratios rather than R^2 values for the baselines.

These issues do not necessarily invalidate the core methodology, but they render the specific performance claims and comparative conclusions untrustworthy in their current form. The authors must replace the non-existent baselines with real, verifiable models and ensure every specific numerical claim is directly supported by a cited table, figure, or supplementary data.

Action items.

- [**writing**] The paper makes several high-impact claims regarding state-of-the-art performance and reasoning capabilities across biology, chemistry, and materials science. While the internal logic of the proposed method is sound, there are significant issues with the factual accuracy of the cited baselines and the traceability of specific quantitative claims. The most critical issue is the citation of non-existent models. The paper repeatedly compares SciReasoner against “GPT-5.5”, “Opus-4.7”, “DeepSeek-V4-P

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 contains significant discrepancies between the caption and the visual content in panel (B), where the described DNA-binding analysis is not shown. Additionally, the captions for panels (A) and (C) are grammatically incomplete or truncated.

Figure 2

Figure 2 effectively visualizes retrosynthesis performance and reasoning traces, but the caption is truncated and grammatically incomplete. Additionally, key axes in panels (A) and (D) lack explicit labels, forcing reliance on the text for metric definitions.

Figure 3

Figure 3 effectively visualizes material property performance and latent space clustering, but the caption for panel (C) is grammatically incomplete. Additionally, the parity plots in panel (C) introduce external model names in the legend that are not defined in the caption.

Figure 4

Figure 4 effectively demonstrates the benefits of structural information through ablation benchmarks and reasoning traces, but the caption for panel (B) misidentifies the dimensionality reduction technique (PCA vs. t-SNE/UMAP) and the caption for panel (C) is cut off.

Figure 5

Figure 5 effectively visualizes the training dynamics and reasoning framework, but the provided caption is truncated, omitting the description for panel (C). Additionally, panel (D) displays a legend entry for a model that does not appear in the chart, and the y-axis label in panel (C) is illegible.

Action items.

- **[science]** Figure 1: The caption describes panel (B) as 'Attention analysis for DNA-binding Gene Ontology prediction' with residues enriched at DNA-binding sites, but the panel displays attention maps for proteins P25144, P70696, and Q9HUS3 (which are not DNA-binding proteins) and lacks the described structural interface visualization.

- [**writing**] Figure 1: The caption for panel (C) is truncated ('Rewards increase after an initial expl'), failing to describe the full content of the reinforcement-learning trajectories shown.
- [**writing**] Figure 1: The caption for panel (A) contains a sentence fragment ('shows the largest gains...') that lacks a subject, making it unclear which method is being referenced.
- [**writing**] Figure 2: The caption is truncated at the end of the description for panel (C) ('(C [joint-opus.pdf]'), failing to describe the content of the large panel showing molecular structures.
- [**writing**] Figure 2: The caption text for panel (A) is grammatically fragmented ('Across template-based... reaches 0.72'), lacking a clear subject to identify the method achieving the score.
- [**science**] Figure 2: Panel (A) displays a bar chart of 'Exact Match' scores but lacks a y-axis label, relying solely on the caption to define the metric.
- [**science**] Figure 2: Panel (D) contains a scatter plot with axes ranging from 0.4 to 1.0 and 0 to 20 respectively, but neither axis has a label or unit definition.
- [**writing**] Figure 3: The caption for panel (C) is truncated mid-sentence ('demonstrate the') and lacks the final period.
- [**science**] Figure 3: Panel (C) parity plots display metrics for 'DeepSeek-v4-pro' and 'GPT-5' in the legend, but the caption only discusses the model's performance without identifying these external baselines.
- [**writing**] Figure 3: Panel (A) y-axis label 'MAE' is repeated for every subplot; a single global label or removal of redundant labels would reduce clutter.
- [**writing**] Figure 4: The caption for panel (B) refers to 'PCA' (Principal Component Analysis), but the plot axes are explicitly labeled 'Component 1' and 'Component 2' with a range of -75 to 50, which is characteristic of t-SNE or UMAP embeddings rather than standard PCA; the caption should be updated to match the visualization method or the plot should be relabeled.
- [**writing**] Figure 4: The caption for panel (C) is truncated mid-sentence ('...with structur'), failing to complete the description of the structural reasoning comparison.
- [**writing**] Figure 5: The provided caption is truncated at the end ('...sampling ef'), cutting off the description for panel (C) which is clearly visible in the image.
- [**writing**] Figure 5: Panel (C) contains a bar chart with a y-axis label ('Share of case-judgements (%)') that is rotated 90 degrees and illegible due to low resolution.
- [**writing**] Figure 5: Panel (D) includes a legend for 'DeepSeek-V4-Pro' (dark blue bar) that is not present in the actual bar chart data, creating a discrepancy between the legend and the plot.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper introduces a novel multimodal foundation model, SciReasoner, which relies heavily on a specific set of domain-native structural representations and training protocols. While the core concepts of "structure-property relationships" and "autoregressive reasoning" are accessible to an adjacent-field PhD, the manuscript frequently introduces specific acronyms, symbols, and named methods without immediate definition, creating barriers for readers not deeply embedded in this specific sub-ecosystem.

Specifically, the term "CoT" (chain-of-thought) is used in Section 1 without the standard parenthetical expansion at the point of first use, despite the full phrase being present. More critically, the mathematical notation in Section 3 (Method) introduces symbols like X_v, $\mathbf{H}_v$, and Θ with their components (θ_{vocab}, etc.) without explicitly defining their semantic role or dimensional constraints in the text surrounding the equations. A reader must infer that X_v is a sequence of tokens and $\mathbf{H}_v$ is the resulting embedding matrix, which is a minor but unnecessary cognitive load.

Furthermore, several domain-specific encoders and representations are referenced by name only: "3Di" (Foldseek's structural alphabet), "SLICES" (crystal representation), "ConfSeq" (molecular conformer representation), and "DAPO" (the RL algorithm). While these are cited, they are not standard vocabulary across the broader ML or scientific AI community. An adjacent-field researcher (e.g., a computer vision or NLP specialist) would likely not know what "3Di" or "SLICES" stands for or how they function without a brief gloss. The paper assumes these are "well known" within the immediate circle of the authors' collaborators or the specific subfield of structural tokenization, which violates the self-containment requirement for an interdisciplinary audience.

Finally, the use of "SOTA" in the abstract and text is a common shorthand but technically an undefined acronym in a formal review context, though less severe than the others. The primary issue is the density of undefined, field-specific proper nouns (SLICES, ConfSeq, 3Di) and symbols (X_v, Θ) that act as "black boxes" for the reader. Defining these at first use would significantly improve the paper's accessibility without diluting its technical precision.

Action items.

- **[writing]** Section 1 (Results), paragraph 5: The acronym 'CoT' is used in 'chain-of-thought (CoT) strategy' without prior expansion. While 'chain-of-thought' is spelled out, the acronym 'CoT' appears immediately after without a standard '(CoT)' marker, and is then used later in the text (e.g., Section 3, Post-training) as a standalone term. Ensure the acronym is explicitly defined at first use as 'chain-of-thought (CoT)'.

- **[writing]** Section 3 (Method), Subsection 'Offline Structure Encoder': The symbol X_v is introduced in the sentence 'encode S into the structural information sequence X_v' without defining what X_v represents (e.g., a sequence of discrete tokens, a vector, or a string). Define X_v explicitly upon first introduction, e.g., '...into a sequence of discrete structural tokens X_v'.

- [**writing**] Section 3 (Method), Subsection 'Structure-Aware Vocabulary Embedding': The symbol $\mathbf{H}_v$ is introduced in Equation 1 without a preceding textual definition of its dimensions or semantic meaning (e.g., 'where $\mathbf{H}_v \in \mathbb{R}^{L_v imes d_{LLM}}$ represents the embedded structural sequence'). Add a clause defining $\mathbf{H}_v$ immediately before or after the equation.
- [**writing**] Section 3 (Method), Subsection 'Pretraining': The symbol Θ is defined as the complete parameter set, but the components $heta_{vocab}, heta_{emb}, heta_{head}, heta_{backbone}$ are introduced without defining what specific layers or modules they correspond to in the Qwen architecture. Briefly map these symbols to the model components (e.g., 'where $heta_{backbone}$ denotes the transformer weights of the Qwen backbone').
- [**writing**] Section 3 (Method), Subsection 'Reinforcement learning': The term 'DAPO' is used in 'Model training is performed with DAPO' without being defined or expanded. While it is cited, a competent adjacent-field reader may not know this specific RL algorithm. Expand the acronym at first use (e.g., 'DAPO (Dynamic Advantage Policy Optimization)') or provide a one-sentence gloss of its function.
- [**writing**] Section 1 (Results), paragraph 2: The term '3Di' is used in 'Foldseek 3Di tokens' without definition. While Foldseek is a known tool, '3Di' (3D-Index) is a specific structural alphabet notation. Define it briefly upon first use (e.g., 'Foldseek 3Di (3D-index) tokens').
- [**writing**] Section 1 (Results), paragraph 4: The term 'SLICES' is used in 'SLICES for crystals' without definition. This appears to be a specific crystal representation method. Provide a brief gloss or expansion at first use (e.g., 'SLICES (Symmetry-Local Invariant Crystal Embedding System) or similar').
- [**writing**] Section 1 (Results), paragraph 5: The term 'ConfSeq' is used in 'ConfSeq tokenizer' without definition. Define this specific 3D molecular representation format at first use.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper's argument structure is largely sound. Conclusions generally follow from premises, and numerical claims in the text align with figure captions and tables. The core claim—that native structural reasoning improves performance in low-homology regimes—is supported by ablation studies and attention analyses.

However, a minor logical gap exists in the Abstract and Introduction regarding the baseline for the F_max improvement. The Abstract states: "increasing F_max from 0.42 to 0.55." While

Section 1.2.1 clarifies that 0.42 corresponds to the ESM2 baseline, the summary sections do not explicitly name this model. This creates ambiguity where the baseline is not immediately entailed by the text alone. To ensure strict logical clarity, the Abstract and Introduction should explicitly state "from 0.42 (ESM2) to 0.55," matching the explicit comparison in the Results section.

No other contradictions, non-sequiturs, or definition drifts were found. Ablation conclusions match tables, causal claims are supported by described interventions, and expert evaluation statistics are consistent.

Action items.

- **[writing]** Abstract and Intro state 'F_max from 0.42 to 0.55' without naming the 0.42 baseline. Section 1.2.1 clarifies 0.42 is ESM2. Add '(ESM2)' after 0.42 in Abstract/Intro to match Results and ensure logical clarity.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper presents a compelling architecture and strong results on specific benchmarks, but the rhetoric frequently exceeds the scope of the demonstrated evidence, particularly regarding the universality of the "interdisciplinary" claim and the completeness of the "solution" to scientific reasoning.

Title and Abstract Overreach: The title "Accurate, Interdisciplinary and Transparent Structure-property Understanding..." and the abstract's claim that the model "solves the joint challenge of representation and reasoning" are too broad. The evidence is restricted to three specific domains (proteins, small molecules, crystals) and a curated set of 86 benchmarks. "Solving" a field-wide challenge implies a general capability not yet demonstrated, and "Interdisciplinary" suggests a breadth (e.g., covering all of biology, chemistry, and materials science) that the specific datasets do not support. The claim should be narrowed to reflect the specific domains tested (e.g., "across proteins, small molecules, and crystals") and the verb "solves" should be replaced with "advances" or "addresses."

Undefined Scope in Results: In Section 2.2.1, the abstract and text claim improvements for "low-homology and orphan-like proteins." While the paper provides specific data for the "low-homology" regime (30% identity), it does not define or provide specific metrics for "orphan-like" proteins. This term is used rhetorically to suggest a broader capability than the data explicitly supports. The authors must either define "orphan-like" with a specific homology threshold and show the corresponding data or remove the reference to orphans to avoid implying a test that was not performed.

Generalization of Modalities: The conclusion asserts that the model unifies reasoning across "major scientific modalities." The experiments cover only three modalities: protein sequences/structures, small molecule graphs/3D structures, and crystal lattices. While these are significant, the phrase "major scientific modalities" implies a universality that includes, for instance, spectroscopy, reaction kinetics, or other biological data types not included in the benchmark suite. The claim should be scoped to "the three tested modalities" to remain faithful to the evidence.

Human Evaluation Certainty: The claim of a "98% preference" in double-blind expert evaluation is presented as a definitive fact. However, the text notes in Section 2.5 that this is a "pilot" with a sample size of N=177 and that "more human judgments" are being collected. Presenting a point estimate from a pilot study as a final, universal statistic without confidence intervals or acknowledging the preliminary nature of the data overstates the robustness of the finding. A more accurate framing would acknowledge the pilot status or provide a confidence interval.

These issues are primarily rhetorical and can be resolved by tightening the language in the title, abstract, and conclusion to match the specific boundaries of the experimental validation.

Action items.

- [**writing**] Title/Abstract claim 'solves' the joint challenge and 'Interdisciplinary' understanding. Results only cover proteins, small molecules, and crystals on specific benchmarks. Replace 'solves' with 'advances' and qualify 'Interdisciplinary' to 'across the tested domains'.
- [**writing**] Abstract claims improvements for 'orphan-like' proteins, but Section 2.2.1 only quantifies gains for 'low-homology' (30% identity). Define 'orphan-like' with a specific threshold or remove the claim as it lacks explicit evidence in the results.
- [**writing**] Conclusion states the model unifies reasoning across 'major scientific modalities,' but only three (proteins, molecules, crystals) were tested. Narrow the claim to 'across the three tested modalities' to avoid implying untested domains like spectroscopy or kinetics.
- [**writing**] Abstract cites '98% preference' in expert evaluation as a definitive fact, but Section 2.5 calls it a 'pilot' (N=177) with more data pending. Qualify the claim as 'in this pilot study' or add confidence intervals to avoid overstating certainty.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

This paper presents a multimodal foundation model (SciReasoner) for scientific reasoning across proteins, small molecules, and inorganic crystals. The work focuses on structure-property relationships, retrosynthesis planning, and functional annotation.

From a safety and ethics perspective, the research is low-risk. The model is designed to assist in scientific discovery (e.g., drug discovery, materials science) and does not inherently possess capabilities for generating harmful biological agents, cyberattacks, or deceptive content. The "dual-use" potential of accelerating drug discovery is a standard characteristic of the field and is not a specific failure of this paper to disclose, as the paper does not claim to generate novel toxins or pathogens, nor does it provide operational instructions for synthesizing hazardous compounds.

The data sources described (UniProt, PDB, Materials Project, USPTO, etc.) are standard public scientific repositories. The paper explicitly states that it excludes data with high sequence identity to test sets to prevent leakage, and it uses public datasets with established licenses for research. There is no indication of scraping data in violation of Terms of Service, nor is there any release of Personally Identifiable Information (PII) or sensitive human subject data. The human evaluation described involves domain experts rating model outputs on scientific plausibility; this is a standard methodological step in AI research and does not constitute a human-subjects study requiring IRB approval in the context of collecting private data or behavioral manipulation.

The paper does not contain any operational details for biohazard synthesis or cyber-offense methods. The reasoning traces provided in the examples (e.g., retrosynthesis of standard organic molecules, protein function prediction) are benign and educational. There are no undisclosed conflicts of interest evident in the text, and the authors acknowledge the use of public data and standard baselines.

Consequently, there are no foreseeable, non-trivial risks of harm that the paper fails to acknowledge, disclose, or mitigate. The work adheres to standard norms for scientific AI research.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling architecture for native structural reasoning, but the experimental design in several key sections relies on single-point metrics and asymmetric comparisons that leave the central claims vulnerable to alternative explanations like sampling noise or compute advantages.

First, the headline performance gains (e.g., 0.72 vs 0.63 in retrosynthesis, 0.55 vs 0.42 in GO prediction) are reported as single numbers without any indication of variance. In deep learning benchmarks, especially with large models, a 1-2 point gap can easily arise from a single lucky random seed or specific test-set ordering. Without reporting results across multiple seeds (e.g., 3-5) with standard deviations, the reader cannot distinguish a robust architectural effect from statistical noise. The claim that SciReasoner is "state-of-the-art" is currently unsupported by evidence of stability.

Second, the comparison in the retrosynthesis task (Fig 2A) appears to confound model capability with inference strategy. SciReasoner is evaluated using 16 stochastic samples with a frequency-based ranking heuristic, while the baselines (RSGPT, Opus-4.7) are not explicitly described as receiving the same sampling budget or selection mechanism. If the baselines were evaluated with a single greedy pass, the reported improvement may simply reflect the benefit of "best-of-N" sampling rather than the model's native reasoning ability. A fair comparison requires holding the sampling budget and selection criteria constant across all models.

Third, the ablation study (Fig 4) attributes performance drops to the removal of structural tokens, but the experimental setup does not clearly isolate this variable. It is unclear if the "structure-free" model was trained with the exact same data mixture, compute budget, and hyperparameters as the full model, or if the removal of structural data led to a reduction in effective training signal. If the control run differs in data volume or training steps, the performance gap cannot be definitively attributed to the structural tokens. A rigorous ablation requires a control run that matches the full model in every way except the presence of the structural modality.

Finally, the human evaluation (Fig 3E) reports a striking 98% preference rate based on 177 judgments. However, the methodology for selecting these 177 cases is opaque. If the cases were selected post-hoc to maximize the observed gap (e.g., by excluding cases where the baseline performed well), the result is biased. The authors should disclose the total pool of evaluated cases, the selection criteria, and the distribution of difficulty levels to ensure the preference rate is representative of general performance.

Action items.

- [**writing**] The paper presents a compelling architecture for native structural reasoning, but the experimental design in several key sections relies on single-point metrics and asymmetric comparisons that leave the central claims vulnerable to alternative explanations like sampling noise or compute advantages. First, the headline performance gains (e.g., 0.72 vs 0.63 in retrosynthesis, 0.55 vs 0.42 in GO prediction) are reported as single numbers without any indication of variance. In deep learning benchmar

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in this paper is generally clear but lacks necessary uncertainty quantification and multiple-comparison controls in specific high-impact sections.

In Section 2.2.1, the authors present reasoning quality scores stratified by BLAST similarity bins (n=20 per bin) with error bars labeled as standard error of the mean (s.e.m.). While the visual representation suggests stability, the text asserts that scores "remain stable" and are "highest" without specifying the statistical test used to compare these bins. With a sample size of only 20 per bin and the likely non-normal distribution of LLM-judge scores (bounded 1–10), the validity of parametric assumptions for the mean is questionable. The authors should report the standard deviation (SD) in addition to the SEM to allow readers to assess the spread of the data and explicitly state whether a non-parametric test (e.g., Kruskal-Wallis or Mann-Whitney U) was used to validate claims of difference or stability across bins.

In Section 2.2.5, the human expert evaluation claims that "Every per-axis difference is significant... P<0.001" based on a Wilcoxon signed-rank test. The study evaluates five distinct quality axes (Q1–Q5) on the same set of 177 paired judgments. Performing five hypothesis tests on the same data without correcting for multiple comparisons inflates the family-wise error rate. Even with a strict alpha of 0.001, the probability of at least one false positive across five tests is non-negligible. The authors should apply a correction method (such as Bonferroni or Holm-Bonferroni) to the p-values or explicitly state that the uncorrected p-values are reported, adjusting the language to reflect the uncorrected nature of the significance claims.

Finally, in Section 2.2.2, the retrosynthesis accuracy of 0.72 is reported as a single point estimate. The methodology describes sampling 16 completions per query and selecting the best, but it does not report the variance of this metric across the test set or across different random seeds for the model training/inference. In deep learning benchmarks, reporting a single number without a standard deviation or confidence interval (e.g., "0.72 ± 0.01 over 3 seeds") makes it impossible to judge if the improvement over the baseline (0.63) is robust or a result of a lucky run. The authors should report the standard deviation of the accuracy metric or the range of results if only a single seed was used.

Action items.

- [**writing**] Section 2.2.1 reports reasoning quality scores stratified by BLAST similarity with error bars labeled 's.e.m.' for n=20 proteins per bin, but the text does not state whether the underlying distribution of scores is normal or if a non-parametric test was used to compare bins. Given the small n and potential skew in LLM-judge scores, report the standard deviation (SD) alongside the SEM and specify the statistical test used for any 'stable' or 'higher' claims across bins.
- [**writing**] Section 2.2.5 (Human Expert Evaluation) claims 'Every per-axis difference is significant... P<0.001' based on a Wilcoxon signed-rank test on N=177 paired judgments. However, the paper reports 5 distinct quality axes (Q1–Q5) tested on the same 177 pairs without mentioning a correction for multiple comparisons (e.g., Bonferroni or Holm). With 5 tests, the family-wise error rate is inflated; apply a correction or report adjusted p-values to support the 'significant' claim for all axes.
- [**writing**] Section 2.2.2 reports Retrosynthesis USPTO-50K accuracy as a single point estimate (0.72) derived from 16 stochastic completions per query, ranked by frequency. The paper does not report the standard deviation or confidence interval of this accuracy metric across the test set, nor does it clarify if the 0.72 is a mean over multiple seeds or a single run. Report the standard deviation of the accuracy metric (or the range if only one seed was used) to quantify the stability of this result.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a complex, multimodal model, and the writing generally succeeds in conveying the high-level architecture and results. However, several specific instances of grammatical friction and structural awkwardness interrupt the reader's flow, requiring minor revisions to ensure the prose is as precise as the science.

The most significant readability issue occurs in Section 2.2.1 (Protein GO prediction). The third paragraph contains a comma splice that joins three independent clauses without a conjunction: "whereas BLAST relies on local sequence similarity, ESM2 encodes evolutionary and sequence-context patterns without explicit structural grounding, \projName{} predicts GO terms..." This forces the reader to pause and re-parse the sentence structure to understand the intended contrast. Splitting this into two sentences or using a semicolon before the final clause would resolve the ambiguity and improve the rhythm.

In Section 2.2.2 (Retrosynthesis), the second paragraph includes a minor but distracting punctuation error: "e.g.\ the ester C–O bond..." The backslash-space before the period is a LaTeX artifact that should be cleaned to standard English punctuation ("e.g., the ester C–O bond..."). Additionally, the list of examples lacks a parallel structure that would make the enumeration clearer.

Section 2.2.4 (Materials) suffers from a lack of signposting between sentences. The transition from the general statement about chemical identity to the specific observation about polymorphs ("Within each compositional domain...") feels abrupt. Adding a simple transition word like "Furthermore" or "Specifically" would guide the reader more smoothly through the logical progression of the argument.

In the Method section, the "Data Source and Processing" subsection relies heavily on passive voice ("protein sequences were associated with..."), which obscures the agency of the authors and makes the text feel distant. Shifting to active voice ("We associated protein sequences with...") would make the methodology more direct and easier to follow. Similarly, the "Post-training" subsection uses "i.e." twice in rapid succession to define the two stages, creating a repetitive and slightly clunky rhythm. Varying the phrasing (e.g., using "namely" for the second instance) would improve the prose quality.

Finally, there are occasional instances of wordiness, such as the phrase "in order to" where "to" suffices, and minor inconsistencies in how examples are introduced. While these do not prevent understanding, addressing them will elevate the manuscript from "clear" to "polished," ensuring the reader can focus entirely on the scientific contribution without being slowed by syntactic friction.

Action items.

- [**writing**] The manuscript presents a complex, multimodal model, and the writing generally succeeds in conveying the high-level architecture and results. However, several specific instances of grammatical friction and structural awkwardness interrupt the reader's flow, requiring minor revisions to ensure the prose is as precise as the science. The most significant readability issue occurs in Section 2.2.1 (Protein GO prediction). The third paragraph contains a comma splice that joins three independent claus